Note: Problems 1–3 are "fundamental" theorems for the real numbers. As such, you should take special care to *only* use the axioms on the "Number System Axioms" handout and results that we have proved in class.

Problem 1. Prove that for all $a \in \mathbb{R}$, we have $0 \cdot a = 0$.

Problem 2. Prove that for all $a \in \mathbb{R}$, we have $-a = (-1)a$.

Problem 3. Prove that for all $a, b \in \mathbb{R}$, we have $(-a)(-b) = ab$.

Problem 4. For each of the following statements, determine whether it is true or false. If it is true, write a *complete and formal* proof. If it is false, provide a counterexample (this should also be written in the same style).

1. For all $a, b, c \in \mathbb{Z}$ with $a \neq 0$, if $a \mid b$, then $a \mid bc$.
2. For all $a, b, c \in \mathbb{Z}$ with $a \neq 0$, if $a \mid (b-1)$ and $a \mid (c-1)$, then $a \mid (bc-1)$.
3. For all $a, b, c \in \mathbb{Z}$ with $a \neq 0$, if $a \mid bc$, then $a \mid b$ or $a \mid c$.

Problem 5. Either prove the following statement is true or provide a counterexample to show that it is false:
For every $a, b \in \mathbb{Z}$, if $ab \equiv 0 \pmod 6$, then $a \equiv 0 \pmod 6$ or $b \equiv 0 \pmod 6$.

Problem 6. Let n be a positive integer. Prove the following statements about congruence modulo n. (Together, these prove that congruence modulo n is an *equivalence relation*.)

1. For all $a \in \mathbb{Z}$, $a \equiv a \pmod n$. (*Reflexivity*)
2. For all $a, b \in \mathbb{Z}$, if $a \equiv b \pmod n$ then $b \equiv a \pmod n$. (*Symmetry*)
3. For all $a, b, c \in \mathbb{Z}$, if $a \equiv b \pmod n$ and $b \equiv c \pmod n$, then $a \equiv c \pmod n$. (*Transitivity*)

Problem 7. Complete Exercise 19 parts (a), (b), and (c) on pages 100 and 101 of the textbook (*Mathematical Reasoning* by Sundstrom).

Problem 8. Do not turn these in, but I recommend completing the progress checks in sections 3.1 and 3.2 in your textbook.